THE SPECTRUM OF COMET IKEYA-SEKI (1965f)*

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ABSTRACT

At solar distances greater than 0.4 a u. before and after perihelion, the spectrum of comet Ikeya-Seki (1965f) contained the usual cometary emissions due to CH, CN, C_2, and C_3. On October 22, 1965, 1.6 days after perihelion, emission lines due to [O I], Na I, K I, Ca II, Cr I, Mn I, Fe I, Ni I, Cu I, and CN were present. A curve-of-growth analysis of the Fe I emission lines leads to a formal excitation temperature of 4500° K. Evidence is found for fluorescence in the spectra of Fe I and Ni I. An upper limit of about 2600° K is obtained for the excitation temperature derived from lines of Na I and K I. The apparent abundance ratio K I/Na I is low and the ratio Cu I/Fe I is unaccountably high in the comet relative to the Sun, meteorites, and Earth's crust. In view of numerous uncertainties, smaller abundance anomalies for other elements cannot be regarded as well established. The absence of the Al I resonance lines is attributed to photo-ionization. Lifetimes against photo-ionization may control the dimensions of the region defined by surface brightnesses in the emission lines of neutral metals. Surface-brightness distributions and estimates of absolute intensities for lines of Fe I and Na I lead to volume densities of 180 cm^{-3} and 8000 cm^{-3} for Fe I and Na I, respectively, near the nucleus of the comet. Intensity distributions in the *P*- and *R*-branches of the CN (0,0) band show the mutilating effects due to fluorescence excited by solar radiation. A rotational temperature of 760° K is derived from the CN lines on October 22.

I. INTRODUCTION

Comet Ikeya-Seki (1965f) attracted world-wide attention (see summary by Marsden 1965) because of the opportunities it afforded to study the behavior of a comet during a close approach to the Sun. Metallic emission lines were detected on October 20, 1965 (Universal Time is used throughout this paper), by Thackeray, Feast, and Warner (1966), and preliminary reports of observations made at Haute Provence on October 21 (Dufay, Swings, and Fehrenbach 1965), at Kitt Peak (Livingston, Roddier, Spinrad, Slaughter, and Chapman 1966), and at Sacramento Peak (Curtis 1966) have already appeared.

Spectroscopic observations of the comet were obtained at Lick Observatory by the writer on six days between October 8 and November 3, 1965. Of these greatest interest centers on daytime observations of October 22, about 1.6 days after perihelion passage. A detailed description of these observations is given here in the hope that they will be combined with the observations made elsewhere on October 20 and 21 to provide a description of variations in the physical characteristics of the comet that occurred near perihelion.

II. THE OBSERVATIONS

The comet could not be seen on the slit of the 120-inch coudé spectrograph on October 22 because of the brightness of the daylight sky. The bright central region of the comet (hereinafter referred to as the "nucleus") was located on the slit, or nearly so, by appropriate orientation in the field-viewing system of the coudé spectrograph, where the comet could be seen as a semistellar disk with a diameter of several seconds of arc. The exposures were interrupted at 20-sec intervals by interposing the field-viewing system to correct for drift of the image. The length of the slit was approximately 1′ for all exposures made on October 22. The dispersions in the ultraviolet, visual, and infrared spectral regions were 2.0, 4.1, and 4.1 Å/mm, respectively. Since these three spectral regions were observed separately, there is no assurance that exactly the same portion of the cometary nucleus was on the slit in all cases. This may be of importance in making comparisons of results from different spectrograms for those elements whose emission lines were restricted to a small angular area (see § IV).

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A journal of the Lick observations of comet 1965f is given in Table 1. The first five columns are self-explanatory. The sixth and seventh columns give the position angles of the slit and of the direction to the Sun, respectively. The eighth and ninth columns give the observed radial velocity and that interpolated from Cunningham's ephemeris (Cunningham 1965; Gingerich 1965), respectively. The tenth column contains the heliocentric distance of the comet at the time of observation.

III. THE EMISSION LINES ON OCTOBER 22

The identifications and intensities of the atomic emission lines present on the spectrograms of October 22 are given in Table 2; many of them can be seen in the spectral regions reproduced in Figures 1–3. The list includes definite identifications of [O I], Na I, K I, Ca II, Cr I, Mn I, Fe I, Ni I, and Cu I. Na I is represented by the 3^2S–4^2P^o transitions at λ3302 as well as by the D-lines. The resonance lines of K I are very strong, but the 4^2S–5^2P^o transition at λ4044 is extremely weak. The Ca II emission lines were also extremely weak on October 22 compared to those reported on October 20 and 21, and because of the relatively small positive radial velocity on October 22 (+9 km/sec) they are badly blended with longward emission components of the sky spectrum. The Ca I resonance line was not observed because, in the confusion of trying to observe in all possible wavelength regions, the region λλ4200–5000 regrettably was omitted! Probably the most surprising characteristic of the spectrum is the strength of resonance lines of Cu I. The *absence* of lines of certain elements from Table 2 is discussed in § V, *f*.

The strongest lines were readily found by eyepiece inspection. However, because of the strength of the sky spectrum, weaker features were found only by careful spectrophotometry. The strongest Fe I emission lines have lengths of about 10″, while the length of the slit was about 60″. Therefore, intensity tracings were made of 1 mm (2″.7) strips at (*A*) the edges of the spectrogram which are free of cometary emissions, and at (*B*) along the slit position of the nucleus of the comet. Then the longward wings of the sky absorption features at (*A*) and (*B*) were carefully compared. In this way many cometary emission lines were found that would have been missed by visual inspection. An example of such a case is provided by Cu I λ3273, shown in Figure 4. The intensities of all emission features were obtained by subtracting intensities in (*A*) from those in (*B*). The intensities are in units of 1 equivalent angstrom in the sky continuum. For $\lambda > 3600$ Å the sky continuum was drawn so as to conform to the "100" level in the *Utrecht Atlas*. For $\lambda < 3600$ Å the continuum was drawn arbitrarily. The continuous spectrum of the comet was very weak compared to the sky on October 22, so that in the writer's opinion errors introduced by neglect of the cometary continuum are unimportant compared to other errors. The conversion of the intensities in Table 2 to relative intensities independent of wavelength and to absolute intensities is discussed in § V, *d*, and V, *g*, respectively.

The CN (0,0) band is well marked on all spectrograms of the λ3880 region between October 8 and November 3. On the 2 Å/mm spectrogram of October 22 the individual lines of the *P*-branch are resolved to *P*22 and the (1,1) band head is weakly present. The CN spectrum is discussed in § VI.

IV. SURFACE DISTRIBUTIONS OF EMISSION INTENSITIES AND RADIAL VELOCITIES

a) *Emission Intensities*

Intensity distributions over the surface of the nuclear region of the comet on October 22 have been determined from series of intensity tracings at 1″.36 or 2″.72 intervals perpendicular to the dispersion for the following lines: Na I λ5890 and λ5896, K I λ7665 and λ7699, Ca II λ3933, Fe I λ3719 and λ4045, and the total intensity of the CN λ3883 *P*-branch from *P*21-*P*24. The results are given in Table 3 and Figure 5. The lines of K I, CN, and probably Ca II have half-intensity lengths of about 10″, while the

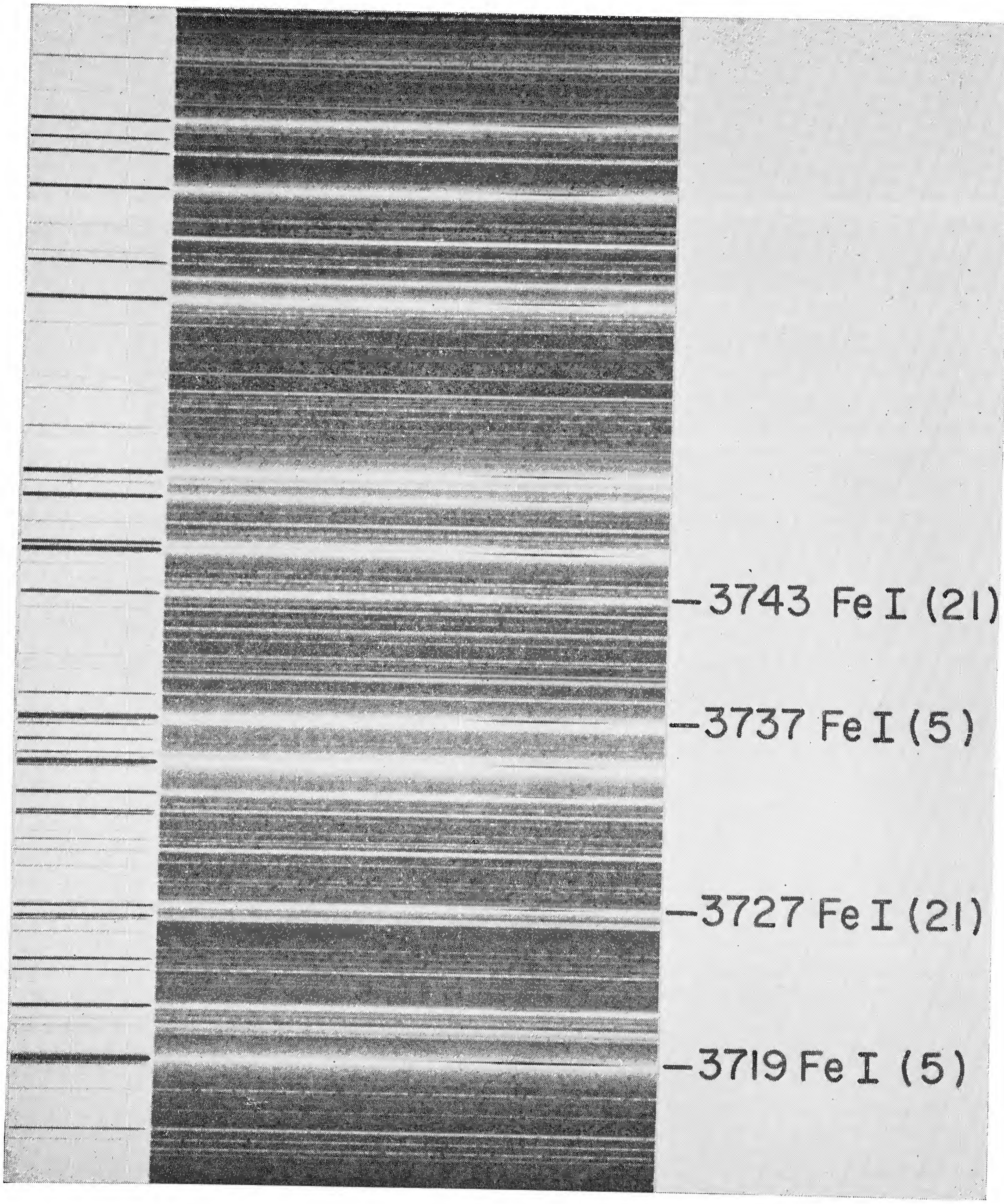

FIG. 1.—The spectral region near λ3750 in comet Ikeya-Seki on October 22.71. Numerous cometary emission lines due to Fe I, RMT multiplets 5 and 21, are present. They are displaced by +9 km/sec relative to the spectrum of the daylight sky. The excitation effects discussed in § V of the text are clearly seen in the ratios λ3719/λ3727 and λ3737/λ3743. The *gf* values for the lines in multiplet 21 exceed those in multiplet 5 for both ratios. The original dispersion was 2.0 Å/mm.

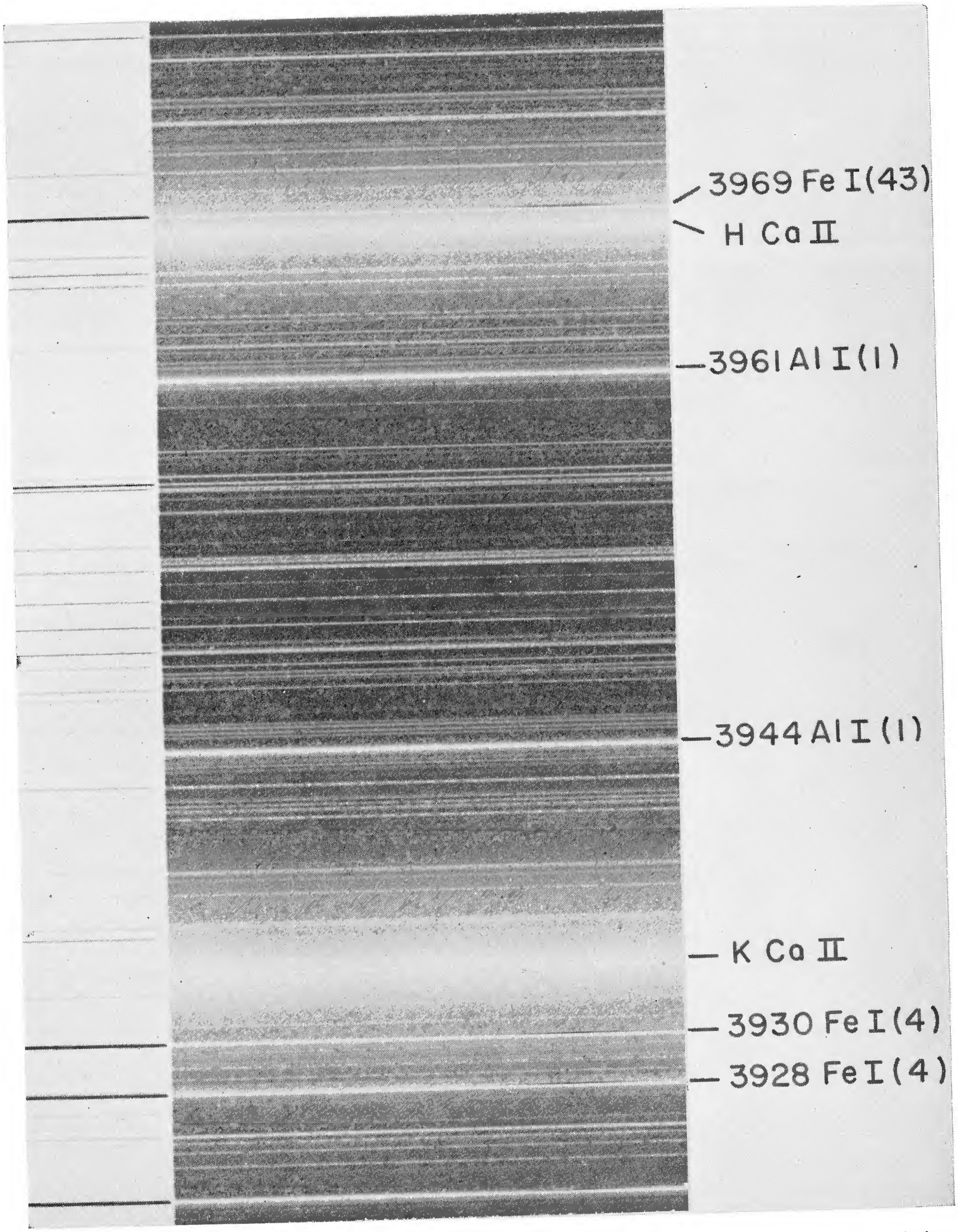

FIG. 2.—The spectral region near Ca II H and K in comet Ikeya-Seki on October 22.71. The emission lines of Fe I multiplet 4 are strong in the comet spectrum while the strong resonance lines of Al I are absent. Note the weakness of the Ca II emission lines relative to their strengths illustrated by Dufay *et al.* (1965) on October 21.44 and by Thackeray *et al.* (1966) on October 20.40. The original dispersion was 2.0 Å/mm.

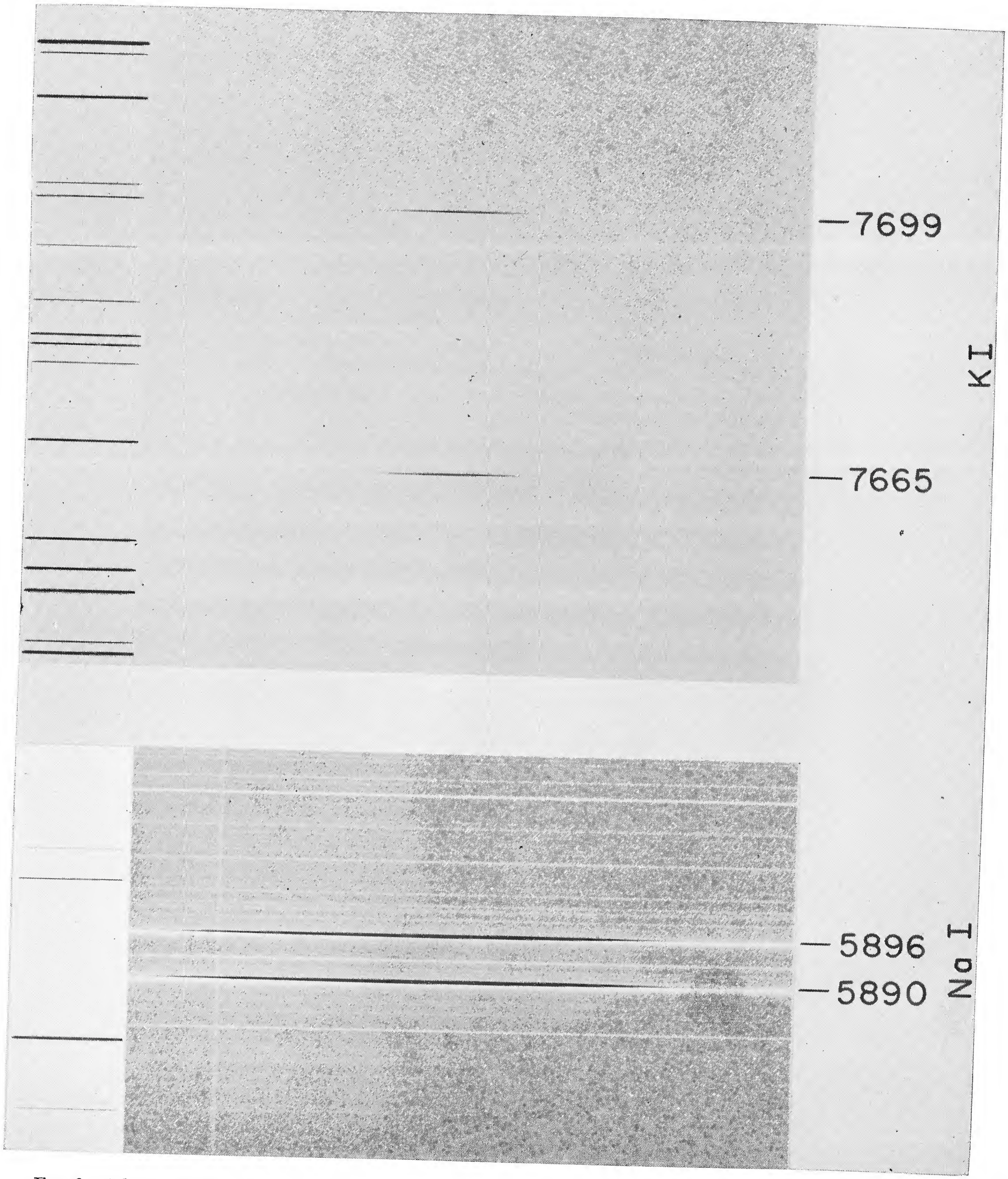

FIG. 3.—The spectral regions near the resonance lines of Na I and K I in comet Ikeya-Seki on October 22.73. The total length of the slit, reproduced here, is 1′. The original dispersion was 4.1 Å/mm in both cases.

TABLE 1

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Plate No.*	U.T. of Midexposure 1965	Exposure	Emulsion	Dispersion (Å/mm)	θ(Slit)	θ(☉)	v_{rad} (km/sec)	$d\Delta/dt$ (km/sec)	r (a.u.)	Emission Lines Present Due to	Remarks†
EC 4617	Oct. 8.560	21^m	IIaO	16.2	116°	76°	−39 4	−36.9	0.59	CH, CN, C_2, C_3	(1)
EC 4624.	9.562	21^m	IIaO	8.1	116°	76°			.56	CH, CN	(2)
EC 4663.	22.709	3^m	IIaO	2.0	149°	88°	+ 9.2	+ 9.1	14	Na I, K I, Ca II, Cr I, Mn I, Fe I, Ni I, Cu I	(3)
EC 4664.	22.727	19^m	I-N	16.2	154°	88°				K I	(3), (4)
EC 4665.	22.744	3^m	103aF	4.1	160°	88°	+10.7	+ 9 0	.14	[O I], Na I	(3)
EC 4666.	22.761	24^m	I-N	4.1	166°	88°				K I	(3)
EC 4667	22.847	17^m	I-N	4.1	197°	88°				K I	(3)
EC 4668.	22.864	1^m	103aF	4.1	203°	88°	+10.1	+ 8.6	15	Na I	(3)
CS 5507	28.53	$3^s,6^s,60^s$	IIaO	430(Hγ)	90°	83°			.40	CN, C_2	(1)
CS 5510.	29.53	$3^s,6^s,60^s$	IIaO	430(Hγ)	90°	83°			44	CN, C_2	(1)
EC 4680..	Nov. 3.565	33^m	IIaO	16.2	122°	83°	− 0.7	+ 1.1	0.63	CH, CN, C_2, C_3	

* The EC series are 120-inch coudé, the CS are Crossley prime focus.

† (1) Appreciable sky continuum; (2) weak exposure; (3) daytime observation, strong sky spectrum; (4) greatly overexposed.

TABLE 2

INTENSITIES OF ATOMIC EMISSION LINES

λ(Å)	El	RMT	log I*	λ(Å)	El	RMT	log I*
6300 30	[O I]	1F	−1 96	3758 24	Fe I	21	−1 15
				3763 79	Fe I	21	−1 20
3302 34	Na I	2	−1 57	3767 19	Fe I	21	−1 30
3302 94	Na I	2	−1 59	3787 88	Fe I	21	−1 77
5889 95	Na I	1	+2:	3795 00	Fe I	21	−1 82
5895 92	Na I	1	+2:	3799 55	Fe I	21	−1 68
				3812 96	Fe I	22	−1 80
4044 14	K I	3	−2 52	3815 84	Fe I	45	−1 24
7664 91	K I	1	+0 16	3820 43	Fe I	20	−0 97
7698 98	K I	1	+0 21	3824 44	Fe I	4	−1 43
				3825 88	Fe I	20	−1 39
3933 66	Ca II	1	−2 73	3827 82	Fe I	45	−1 72
				3834 22	Fe I	20	−1 34
3578 69	Cr I	4	−2 00	3840 44	Fe I	20	−1 46
3605 33	Cr I	4	−2 40	3841 05	Fe I	45	−1 96
				3849 97	Fe I	20	−1 82
4030 76	Mn I	2	−1 77	3856 37	Fe I	4	−1 29
				3859 91	Fe I	4	−0 97
3440 61	Fe I	6	−1 18	3865 53	Fe I	20	−2 04
3440 99	Fe I	6	−1 44	3878 02	Fe I	20	−2 04
3465 86	Fe I	6	−1 68	3878 58	Fe I	4	−1 49
3475 45	Fe I	6	−1 48	3886 28	Fe I	4	−1 24
3476 70	Fe I	6	−1 89	3887 05	Fe I	20	−1 96
3490 58	Fe I	6	−1 40	3895 66	Fe I	4	−1 62
3497 84	Fe I	6	−2 10	3899 71	Fe I	4	−1 62
3513 82	Fe I	24	−2 15	3902 95	Fe I	45	−1 72
3526 04	Fe I	6	−2 15	3920 26	Fe I	4	−1 82
3558 52	Fe I	24	−1 92	3922 91	Fe I	4	−1 72
3565 38	Fe I	24	−1 30	3927 92	Fe I	4	−1 58
3570 10	Fe I	24	−1 17	3930 30	Fe I	4	−1 47
3581 20	Fe I	23	−1 08	3969 26	Fe I	43	−1 66
3585 32	Fe I	23	−1 89	4045 82	Fe I	43	−0 89
3585 71	Fe I	23	−1 96	4063 60	Fe I	43	−1 33
3618 86	Fe I	23	−1 66	4071 74	Fe I	43	−1 64
3618 77	Fe I	23	−1 33				
3631 46	Fe I	23	−1 14	3380 57	Ni I	37	−2 04
3647 84	Fe I	23	−1 16	3392 99	Ni I	20	−2 15
3705 57	Fe I	5	−1 40	3414 76	Ni I	19	−2 10
3709 25	Fe I	21	−1 70	3446 26	Ni I	20	−1 82
3719 94	Fe I	5	−0 72	3458 47	Ni I	19	−1 72
3722 56	Fe I	5	−1 70	3461 65	Ni I	17	−2 16
3727 62	Fe I	21	−1 47	3492 96	Ni I	18	−1 46
3733 32	Fe I	5	−1 66	3510 34	Ni I	18	−2 22
3734 87	Fe I	21	−1 26	3515 05	Ni I	19	−1 96
3737 13	Fe I	5	−0 99	3524 54	Ni I	18	−1 46
3743 36	Fe I	21	−1 58	3566 37	Ni I	36	−2 16
3745 56	Fe I	5	−1 06	3619 39	Ni I	35	−1 92
3745 90	Fe I	5	−1 68				
3748 26	Fe I	5	−1 54	3247 54	Cu I	1	−1 24
3749 49	Fe I	21	−1 10	3273 96	Cu I	1	−1 64

* I = equivalent width of the emission line in units of the local continuum

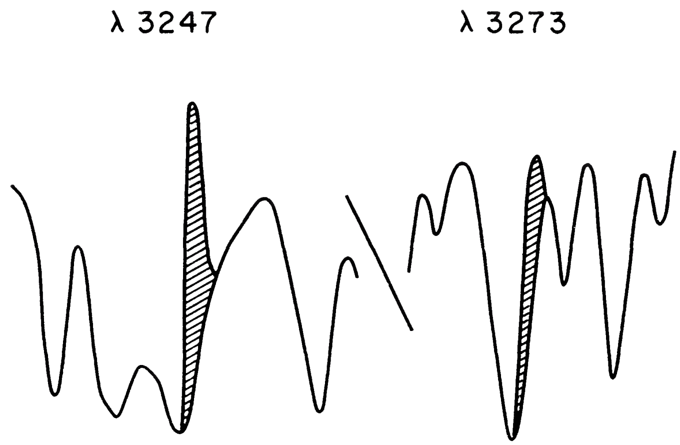

FIG. 4.—Profiles of the resonance lines of Cu I superposed on the spectrum of the daylight sky. The hatched areas represent the cometary emission features. They were derived by the method described in § III of the text.

TABLE 3

SURFACE BRIGHTNESS DISTRIBUTIONS

COORDINATE*	LOG I							
	Na I		K I		Fe I		Ca II† λ3933	CN† P_{21}-P_{24}
	λ5889	λ5895	λ7665	λ7699	λ3719	λ4045		
+21″75	−1 17	−1 34						
+19 05	−0 96	−1 12						
+17.70	−0 74	−0 82						
+16 30	−0 55	−0 72						
+14 95	−0 25	−0 42						
+13 60	+0 07	−0 19						
+10 90	+0 56	+0 57					−3 62	
+ 8 15					−1 95	−2 39	−3 31	0 18
+ 6 80			−0 70	−0 64				
+ 5 45				− 34	−1 72	−2 03	−2 99	46
+ 4 10			− 12	− 12				
+ 2 70			+ 05	+ 10	−1 12	−1 36	−2 80	81
+ 1 35			+ 14	+ 18	−0 96			
0 00			+ 16	+ 21	−0.72	−0 89	−2 73	92
− 1 35			+ 19	+ 19	−0 72			
− 2 70			+ 12	+ 16	−0 82	−1 04	−2 79	89
− 4 10			+ 01	+ 08				
− 5 45			− 18	− 09	−1 38	−1 76	−2 94	58
− 6 80			− 36	− 24	−1 74			
− 8 15			−0 74	− 51	−2 05	−2 35	−3 18	0 20
− 9 55				−0.66				
−10 90							−3 43	
−13 60	+0 66	+0 51						
−16 30	+0 40	+0 16						
−17 70	+0 22	0 00						
−19 05	+0 07	−0 26						
−20 40	−0 26	−0 43						
−21 75	−0 44	−0 68						
−23 10	−0 66	−0 72						
−24 50	−0 89	−1 00						

* Measured perpendicular to the dispersion, with positive value increasing northward

† Smoothed

half-intensity length for Fe I is about 5″. Both exposures of the Na I D-line region are too dense for microphotometry over the central 20″. The dashed interpolation for the Na I lines was obtained by arbitrarily fitting the K I distribution for the central region to the Na I distribution measured at distances greater than 10″ from the nucleus. It should be noted that it also would have been possible to fit the Fe I distribution to that for Na I and that neither the K I nor the Fe I distributions may be appropriate for interpolation through the center of the D-line arcs if self-absorption in the D-lines is appreciable. The ratio $D_2/D_1 \sim 1.4$ in the measurable region suggests that self-absorption

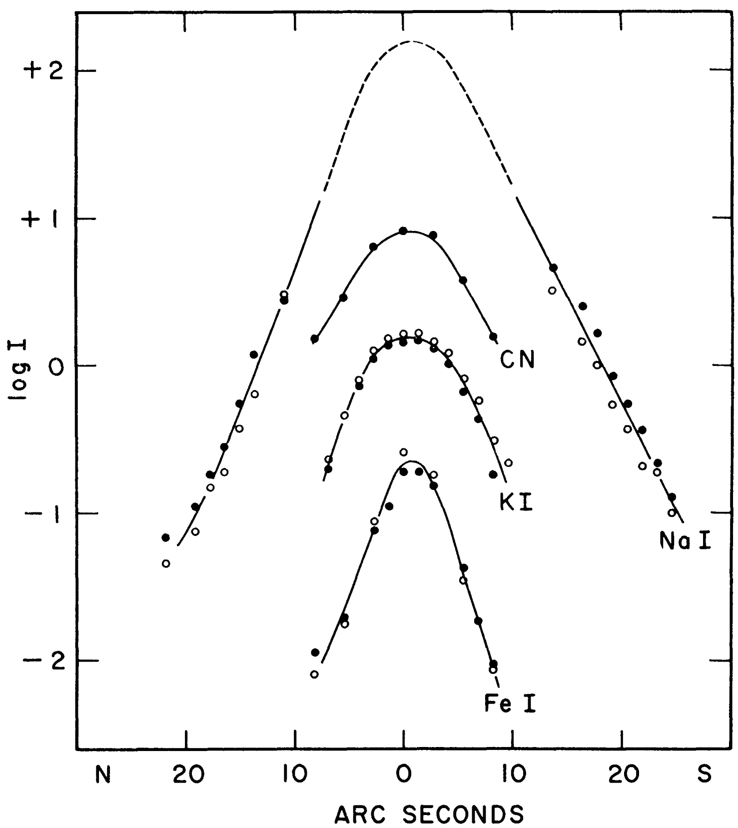

FIG. 5 —Surface brightness versus angular distance from the cometary nucleus for lines of Na I, K I, Fe I, and CN. Filled and open circles denote, respectively, λ5890 and λ5896 for Na I, λ7665 and λ7699 for K I, λ3719 and λ4045 for Fe I The intensity scale for CN is arbitrary. The inversion of the expected intensity ratio of the K I lines is due to telluric O_2 absorption

is important. The intensity ratio for the K I resonance doublet cannot be determined because of the severe interference by telluric O_2 absorption at λ7665.

Since the wings of the Na I and Fe I distributions are very nearly linear in Figure 5, they can be described by an equation of the form

$$I = I_0 e^{-a/a_0} . \tag{1}$$

On the other hand, there is an indication of curvature of the K I distribution which suggests an equation of the form

$$I = I_0 e^{-(a/a_0)^2} . \tag{2}$$

The parameters I_0 and a_0 for these distributions are given in Table 4. Equation (2) fits all of the K I data well. Equation (1) represents the linear portions of the Na I and Fe I distributions, but, of course, fails to reproduce the curvature near the origin. In this respect it is difficult to assess the extent to which atmospheric seeing and guiding errors affect the observed brightness distribution near the nucleus. The K I profiles may indeed be affected by guiding errors during the relatively long infrared exposures (see Table 1). Regardless of the formal representation of the observed brightness distributions, it appears that there are two systems of cometary emissions: (1) a diffuse system represented by K I, CN, and probably Na I, Ca II, and [O I], and (2) a nuclear system defined by Fe I which seems to be representative of Ni I and Cu I as well. This characteristic was noted on October 21 by Dufay *et al.* (1965).

b) *Radial Velocities*

Five spectrograms have been measured for radial velocity. On EC 4617 and EC 4680, obtained on October 8 and November 3, respectively, the seven CN (0,0) lines *R*11–*R*17 were measured on a Grant spectrocomparator. On the basis of measures of four Fe I lines in the sky continuum of EC 4617, evidence was found for a run of about 0.03–0.04 Å in measured wavelength along the length of the slit. The run is probably introduced by a slight difference in the profiles of the upper and lower Fe I comparison lines. This small

TABLE 4

PARAMETERS OF SURFACE BRIGHTNESS DISTRIBUTIONS

ELEMENT	$I=I_0e^{-(a/a_0)}$			$I=I_0e^{-(a/a_0)^2}$
	Fe I	Na I	K I	K I
a_0 (arc sec)	2 1	2.7	2.9	5 6
log I_0 .	−0 40	2 66	0 50	0 14

variation was applied as an empirical correction to the measures of the cometary lines. The average radial velocities from the CN lines, corrected for the diurnal motion of Earth, are given in Table 5. The laboratory wavelengths of Weinard (1955) were used to compute the radial velocities. The sizes of the average deviations indicate that there is no convincing evidence for differential motion across the face of the comet on October 8 or November 3.

Three spectrograms of October 22 have been measured. Spectrogram EC 4663 was measured in two different ways. First, thirty-seven cometary emission lines of Fe I, Ni I, and Cu I in the interval λλ3240–3650 were measured at the position of the nucleus relative to a system of unblended solar lines whose wavelengths were taken from the *Revised Rowland's Table of Wavelengths*. A correlation, shown in Figure 6, was noticed between the intensity of a line and its displacement. This correlation can be interpreted in two ways. If the cometary lines were asymmetric with violet wings, then the density centroids of strong lines would be violet-displaced relative to those of weak lines as observed. On the other hand, if symmetrical emissions are superposed on a background that increases steeply with increasing wavelength, then the same effect will be observed. The longward sides of the solar absorption contours on EC 4663 certainly provide such continuum gradients and experimentation with representative profiles shows that the latter effect is entirely capable of explaining the correlation in Figure 6.

In a second measurement of EC 4663, ten strong lines of Fe I were measured at 2″.7 intervals along the slit. The results are given in Table 5. Taken literally they imply a gradient in radial velocity of 0.1 km/sec/(arc sec) from center to edge of the nuclear region defined by the Fe I lines. However, in view of the complications introduced by gradients in the background sky spectrum, the existence of such a velocity gradient must be considered doubtful at present.

Finally, the D-lines were measured on EC 4665 and EC 4668 at 2″7 intervals along the slit. Because of the paucity of solar lines in the sky background near the D-lines, the systematic errors in the D-line velocities are probably greater than those for EC 4663. The results are given in Table 5 and Figure 7. The D-lines are so heavily overexposed along most of the slit (see Fig. 3) that the argument given above involving the background continuum cannot apply. We have measured either a genuine variation of velocity with position across the face of the comet or the effect of an asymmetrical pro-

TABLE 5

RADIAL-VELOCITY DISTRIBUTIONS

Coordinate*	$\langle v_{rad} \rangle \pm \langle \lvert AD \rvert \rangle$ (km/sec)	Coordinate*	v_{rad} (km/sec)	v_{rad} (km/sec)
	EC 4617 (CN)		EC 4665 (Na I)	EC 4668 (Na I)
+20″..	−38.5 ±0.7			
+10 .	−39 2 1 0	+24″6	14.9	....
0..	−39.5 0.5	+21.8	14 1	....
−10 .	−39 5 1.2	+19 1	12 9	11.4
−20..	−38 5 ±1 0	+16 4 .	12.0	12 2
		+13 6	12 4	11 0
	EC 4680 (CN)	+10.9	11 1	10 8
		+ 8 2	11.3	10.6
+22...	− 1 1 ±1 7	+ 5 5	11 1	10 2
+11	− 0.2 0.6	+ 2 7	10 8	9.8
0.	− 1 3 0 7	0.0	10.8	10 7
−11 .	− 0 9 1.3	− 2 7	10 4	10.9
−22 .	− 1.5 ±1 7	− 5.5	11 1	9 6
		− 8 2	11.2	10.7
	EC 4663 (Fe I)†	−10 9	11 6	11 0
		−13 6	11 5	11 2
+5″4.	+ 9 77 ±0 42	−16 4	12 2	11 8
+2 7	+ 9 48 .20	−19 1	12 9	12.4
0 0.	+ 9 17 05	−21 8	13 4	11 8
−2 7	+ 9 48 01	−24 6	14 1	14 2
−5 4.	+ 9 71 .06			
−8 1.... .	+ 9 99 ±0 08			

* Measured perpendicular to the dispersion with positive values increasing northward

† Probable errors, not average deviation, are listed for EC 4663

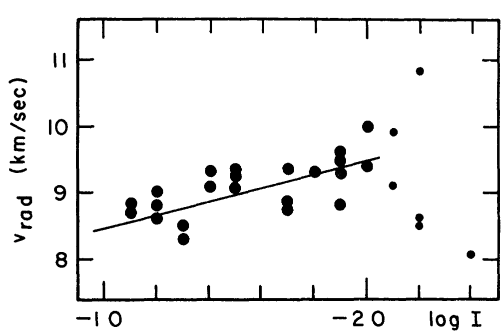

FIG. 6.—The correlation between measured radial velocity and log I for emission lines on spectrogram EC 4663. Small symbols represent the weakest lines, for which velocity measurements are uncertain.

file. Unfortunately, the sky background seriously hampers efforts to measure asymmetry at the tips of the arcs where the photographic density in the lines is low enough to permit accurate photometry.

Radial velocities of the nuclear region for all five spectrograms are given in Table 1 together with the $d\Delta/dt$ interpolated from Cunningham's ephemeris. The average value for EC 4663 was based on all measured lines for which $\log I < -2.0$.

V. ANALYSIS OF THE ATOMIC LINES

a) *The Curve of Growth for Fe* I

A preliminary plot of $\log I$ versus $\log gf$ values from Corliss and Bozman (1962) for the Fe I lines showed immediately that lines of RMT multiplets 4–6, 20–24, and 43–45

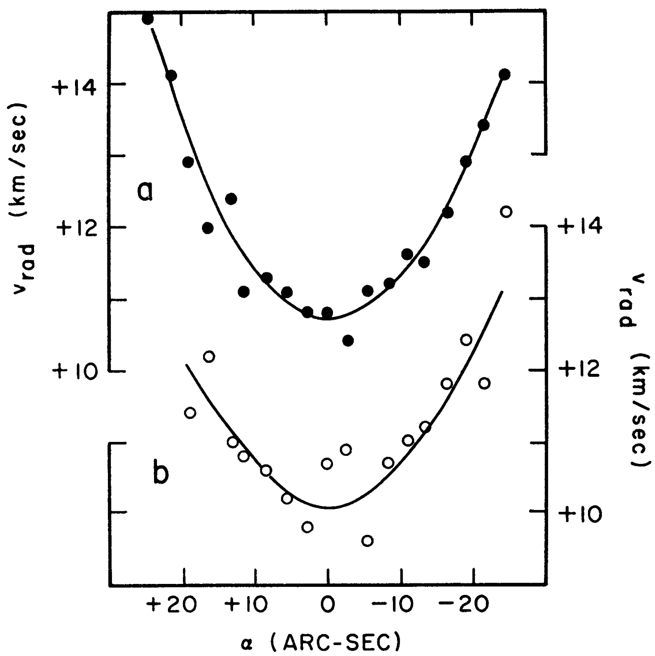

FIG. 7.—Measured radial velocity versus angular distance from the cometary nucleus for the Na I D-lines on (*a*) spectrogram EC 4665 and (*b*) spectrogram EC 4668.

with upper excitation potentials of 3.4, 4.3, and 4.7 eV, respectively, define three separate relations that can be combined by means of a Boltzmann distribution in the excited levels with $\theta \sim 1$. A least-squares solution of

$$\log \frac{\lambda^3 I}{gf} = \text{const.} - \theta\chi \tag{3}$$

gives $\theta = 1.11 \pm 0.05$ or $T_{\text{exc}} = 4500°$ K. The resulting curve of growth is shown in Figure 8. The temperature is unexpectedly high compared to the black-sphere temperature of 800° K at the solar distance of the comet on October 22.7 U.T. (=0.14 a.u.). Since the lower levels of all transitions in RMT multiplets 20–45 are strongly metastable, we suspect that the behavior of Fe I resembles that of C_2, which gives an anomalously high formal "temperature" in comets at all solar distances. In the case of C_2 the steady-state populations in the lower vibrational levels are built up by radiative rather than collisional processes since collisions are rare in cometary gases generally (this is verified for the present comet by the densities derived in § V, *g*), and the rotation spectrum of C_2

is forbidden. The observed emission is ascribed to resonance fluorescence rather than to thermal emission at the temperature implied by the intensity distribution in the rotation-vibration bands.

b) Resonance Fluorescence in the Fe I *Spectrum*

If the Fe I emission spectrum is due to scattering of solar radiation by cometary Fe I atoms, we should expect to find mutilation of multiplets analogous to the well-known effects in CN and C_2 due to the blocking effect of Fraunhofer absorption lines in the solar spectrum (Swings 1941; McKellar 1942, 1944). Inspection of the energy-level diagram

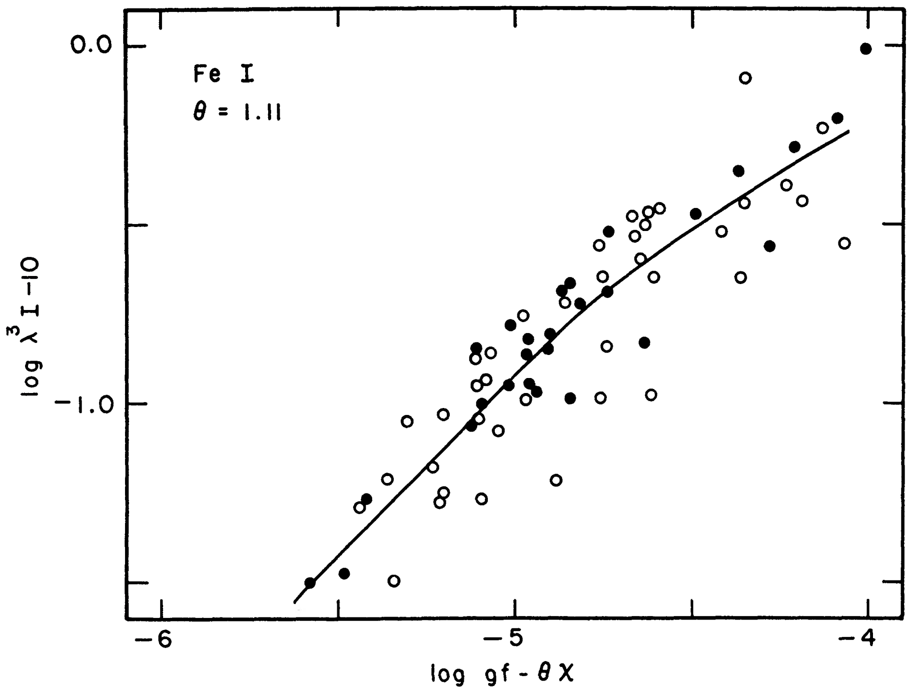

FIG. 8.—Curve of growth for cometary Fe I emission lines Filled and open circles denote lines for RMT multiplets 4–6 and 20–43, respectively The freehand curve through the data was used to estimate the abundances relative to Fe I given in Table 9.

for Fe I shows immediately that the Fe I multiplets under consideration are not, in general, subject to strong mutilation effects because individual upper states communicate in most instances with several lower states including those in the ground term. In such cases the likelihood is small that the upper state of an observed transition will have a greatly abnormal population due to Fraunhofer absorption at all relevant wavelengths. However, a systematic search for suitably fluorescence-sensitive lines led to the list in Table 6. The quality estimate "A" in the third column of the table indicates that the numbers of transitions per second to the upper state from all other lower states are smaller by a factor of 10 or more than the number of transitions per second from the lower state of the transition under consideration. The letters "B" and "C" denote lower quality lines for which the above-mentioned factors are between 5 and 10 and between 2 and 5, respectively.

The radial velocity of the comet relative to the Sun on October 22.7 U.T. was +110 km/sec according to Cunningham's ephemeris. In order to look for mutilation effects and to allow for possible systematic motions of the Fe I gas relative to the nucleus to which the orbital elements refer, the relative intensities $F_\lambda^\odot(v)$ were estimated from the *Utrecht*

Atlas of the solar spectrum, Doppler-shifted by +100, +105, +110, +115, and +120 km/sec, for each of the lines in Table 5. The profiles in the *Utrecht Atlas* refer to the center of the disk. We neglect the broadening effect due to the solar equatorial rotational velocity of 2 km/sec. The values of log $F_{\lambda}^{\odot}$ (v) together with the deviations Δ log $\lambda^3 I$ from the mean curve of growth in Figure 8 are also given for each line in Table 5. Finally, we have plotted the values of Δ log $\lambda^3 I$ versus log $F_{\lambda}^{\odot}$ (v) for the middle three velocity values above in Figure 9. The slopes of the best-fitting straight lines through the resulting diagrams are given at the bottom of Table 6. For the velocity +110 km/sec the slope is +0.82; for the four remaining cases the slope is zero or negative. When it is recalled that, for reasons we have not explored, the populations of the Fe I energy levels resemble those of a Boltzmann distribution at T = 4500° K, the slope of the correlation

TABLE 6

Fe I LINES CHOSEN TO TEST THE FLUORESCENCE HYPOTHESIS

λ(Å)	RMT	QUALITY	Δ LOG $\lambda^3 I$	$-$LOG $F_{\lambda}^{\odot}(v)$				
				v (km/sec)				
				100	105	110	115	120
3565 80 .	24	C	+0 05	0 07	0 46	0 35	0.05	0 05
3570.10 .	24	B	− 05	.35	.33	.46	.15	.07
3581 20	23	A	− 10	.19	19	21	.34	25
3608.86 .	23	B	− .38	.07	.11	46	.19	.09
3618 77..	23	B	− 05	.11	.12	.34	26	55
3631 46 .	23	C	+ 13	24	.09	08	26	15
3719 94.	5	A	+ .20	.12	.13	.14	14	46
3737 13..	5	B	+ 06	.43	36	40	44	47
3859 91..	4	A	+ 06	.42	.38	.39	.38	52
3878 58.	4	C	− .15	.29	14	26	22	09
3895 66..	4	A	+0 18	0 05	0.05	0 06	0 11	0 45
Slope	....	...	.. ..	0 00	−0 25	+0 82	0 00	−0 43

for v = +110 km/sec in Figure 9 differs little from the slope of +1.0 that would be expected if the distribution is a Boltzmann distribution and the emission is due solely to resonance fluorescence. It should be noted that, while we have considered levels whose populations are dominated by single transitions, actually all levels may be affected by solar Fraunhofer absorption lines to a limited extent, if the above interpretation is correct. However, a complete analysis of the Fe I spectrum and of the scatter in Figure 8 is beyond the scope of this report.

c) The Spectrum of Ni I

A curve of growth for Ni I was constructed by means of an *assumed* formal excitation temperature of 4500° K. The large scatter in this diagram led to a search for resonance fluorescence effects in the Ni I spectrum with the surprising result that all of the observed Ni I lines save one (λ3392) are strongly mutilation-sensitive. The Δ log I versus log $F_{\lambda}^{\odot}$ (v = +110) diagram in Figure 10 shows that the Ni I spectrum is almost certainly a fluorescence spectrum. If the mean line in Figure 10 is used to correct the observed intensities to constant log $F_{\lambda}^{\odot}$, the curve of growth in Figure 11 is obtained.

d) The Relative-Intensity Scale for the Emission Lines

In order to compare lines for widely differing wavelength regions, as is required for the alkali metals, it is necessary to consider the variation of the relative intensities with wavelength because all cometary emissions have been expressed in units of the local sky continuum. Initially, corrections were applied to the observed cometary emissions on the assumption that the energy distribution of the sky spectrum was that of Rayleigh-scattered sunlight. Deviations of individual lines from a preliminary mean curve of growth for Fe I lines in the interval λλ3400–4000 were then found to vary systematically with wavelength in the sense that the corrected intensities at shorter wavelengths were too large. When these deviations were removed by means of an empirical correction

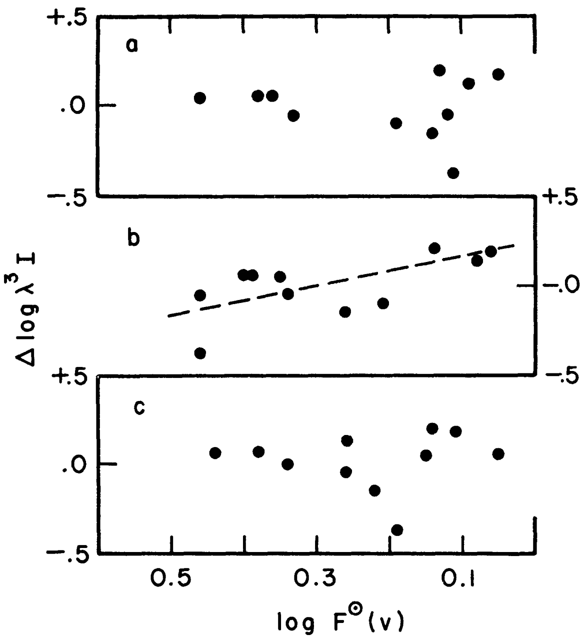

FIG. 9.—The correlation between deviations from the mean curve of growth in Fig 8 and the logarithm of the Doppler-shifted solar flux for (*a*) $v = +105$ km/sec, (*b*) $v = +110$ km/sec, and (*c*) $v = +115$ km/sec.

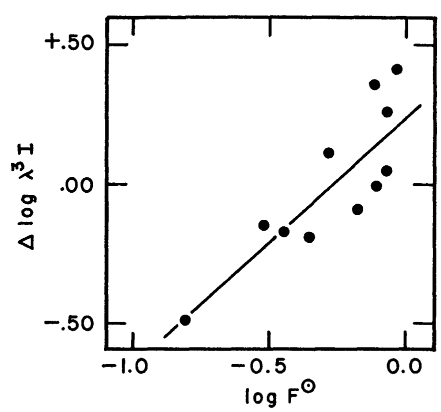

FIG. 10 —The correlation between deviations of Ni I lines from the Ni I curve of growth and the logarithm of the Doppler-shifted solar flux for $v = +110$ km/sec.

curve, it was found that the original measured intensities had very nearly been recovered. This can be understood if it is assumed that there was a predominant neutral component present in the scattered light of the sky in addition to the Rayleigh component. At the angular distance of the comet from the Sun ($\sim 7\overset{\circ}{.}5$), the observations by Abbot (1922) and the compilation of data by van de Hulst (1947) indicate that, in fact, the "white" solar aureole due to aerosols in the Earth's atmosphere is generally appreciable, so that our result for Fe I is not unexpected.

A crude inference about the energy distribution of the sky near the comet on October 22.7 can be obtained as follows. Assume that the sky light contains a Rayleigh component and a neutral component and that the cometary emissions are subject to atmospheric extinction as derived from nighttime observations of stars. For the latter I have used scanner measurements of monochromatic extinction made by Dr. E. J. Wampler on the night of October 22 that preceded the daytime observations of the comet. If the ratio of

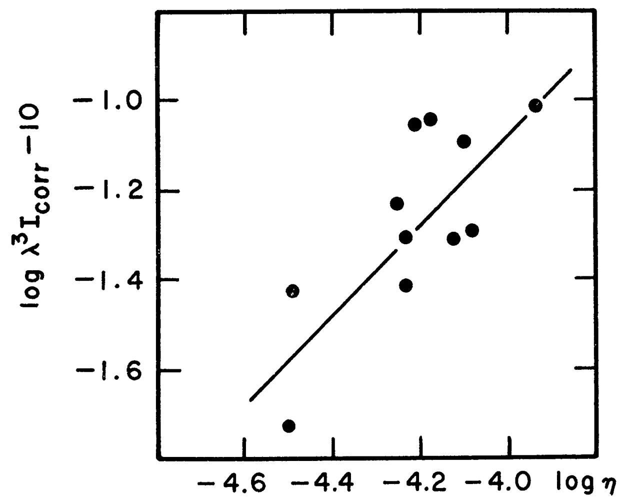

FIG. 11 —The curve of growth derived from Ni I emission intensities corrected to constant solar flux by the slope in Fig. 10. The abscissa is log η = log $gf - \theta\chi$.

the observed intensity to the intensity outside the Earth's atmosphere is $f(\lambda) = I/I_0$, and if the sky brightness is approximated by the expression

$$F_\lambda^{\text{sky}} = F_\lambda^{\odot}\left[a + b\left(\frac{\lambda_0}{\lambda}\right)^4\right], \tag{4}$$

where F_λ^{sky} and $F_\lambda^{\odot}$ are the intensities of the sky and Sun, respectively, and a and b are constants, then the condition that the observed intensities of Fe I lines in units of the continuum be true relative intensities on the interval $\lambda\lambda$3400–4000 implies that

$$f(\lambda) = F_\lambda^{\odot}\left[a + b\left(\frac{\lambda_0}{\lambda}\right)^4\right] \tag{5}$$

on this interval. This condition is satisfied if $a = 0.155$ and $b = 0.019$. If extinction were ignored ($f = 1$ in eq. [5]), then $a = 0.152$ and $b = 0.119$. The predicted energy distribution of the sky and extinction curve used to determine $f(\lambda)$ are given in Figure 12. Also shown is the theoretical distribution plotted by Newkirk (1956) for dielectric particles with an empirical size distribution given by Junge (1953). The writer is not aware of observational data on the specific intensity of the sky spectrum for $\lambda < 4500$ Å. From Figure 12 it appears that there are uncertainties as large as 30 per cent in the correction factors required to convert observed I values to true relative I values for lines in widely separated wavelength regions. This is an unfortunate but unavoidable consequence of the fact that no measurements of sky brightness near the comet were made at Mount Hamilton on October 22, 1965.

e) The Spectra of the Alkali Metals

The cometary emissions of Na I and K I are of particular interest because for both elements transitions from two non-metastable levels to the ground term are observed. Therefore, it is possible to obtain information about the populations of excited states

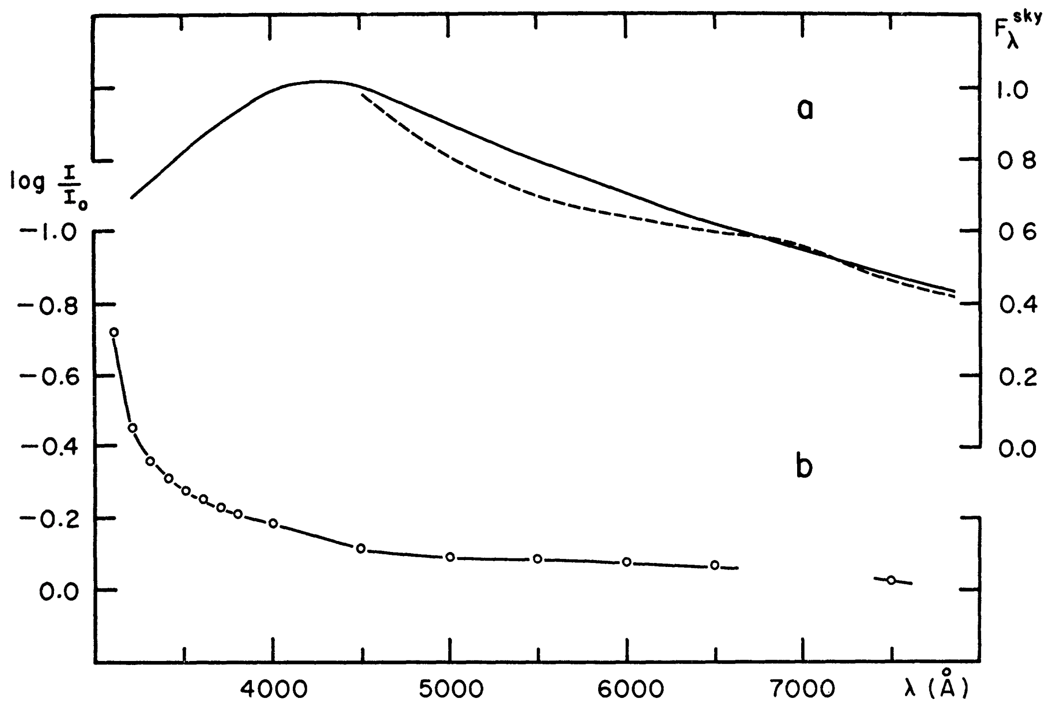

FIG. 12.—(*a*) The solid curve denotes the intensity distribution in the daylight sky 7°5 from the Sun derived by the method in the text. The dashed curve denotes a theoretical distiibution calculated by Newkirk (1956) (*b*) The wavelength dependence of atmospheric extinction on Mount Hamilton on the night of October 22, 1965, determined by Wampler (private communication).

that are free of the complications due to metastability as in the case of Fe I and Ni I. As for Fe I we represent these populations formally by a Boltzmann distribution. If self-absorption is unimportant, then the surface brightness of a region of the comet under consideration in an emission line may be written

$$CI = \frac{1}{4\pi}\left(\frac{8\pi^2 e^2 h}{m}\right)\frac{N_0 g f}{u\lambda^3}\times 10^{-\theta\chi}, \qquad (6)$$

where I is the *relative* intensity, N_0 is the total number of atoms/cm^2 in the line of sight, u is the partition function, C relates relative intensities to absolute intensities, and all other symbols have their usual meanings.

Spectroscopic data for emission lines of the alkali metals are collected in Table 7. The relative intensities I' are corrected for extinction and are given in units of the continuum at λ4000 according to the data in Figure 12. Since the resonance lines of Na I and K I in the comet spectrum were very strong and probably self-absorbed, all that can be learned about the formal "temperature" from equation (6) is that

$$\theta \geq \frac{\log[(\lambda^3 I')_j/(\lambda^3 I')_k] - \log[(gf)_j/(gf)_k]}{\chi_k - \chi_j}, \tag{7}$$

where j and k denote upper levels of lesser and greater excitation, respectively. From the data in Table 7 we thus find that

$$\theta(\text{Na I}) \geq 1.96 \quad \text{and} \quad \theta(\text{K I}) \geq 1.78 .$$

The similarity of the two results suggests that we can adopt for the alkali metals the single lower limit $\theta > 1.9$. For each line of each element an abundance estimate may be obtained from equation (6) in the form

$$\log N = \text{constant} + \log \lambda^3 I' - \log gf + \log u + \theta\chi . \tag{8}$$

TABLE 7

SPECTROSCOPIC DATA FOR ALKALI METALS

Element	λ(Å)	log I	log $\lambda^3 I'$	log gf	χ(upper) (eV)
K I . . .	4044 14 7698 98	−2 52 +0 20	−1 54 +1 52	−0 63 −0 16	3 05 1 60
Na I . .	3302 34 3302 94	−1 58	−0 84	−1 12	3 74
	5889 95 5895 92	+2 10	+3 36	−0 18	2 10
Li I .	6707 8	<−2 0	.	−0 10	1 84

Hence we have

$$\begin{aligned} &\text{from } \lambda 3302\text{: } \log N(\text{Na I}) = 0.64 + 3.74\,\theta + \text{const.}, \\ &\text{from } \lambda 5890\text{: } \log N(\text{Na I}) = 3.86 + 2.10\,\theta + \text{const.}, \\ &\text{from } \lambda 4044\text{: } \log N(\text{K I}) = -0.57 + 3.05\,\theta + \text{const.}, \\ &\text{from } \lambda 7699\text{: } \log N(\text{K I}) = 2.02 + 1.60\,\theta + \text{const.} \end{aligned} \tag{9}$$

The relative abundances of Na I and K I have been estimated by pairing the strong (lower excitation) and weak (higher excitation) transitions separately. In this way the effects of self-absorption are partially taken into account for the strong lines. The results are given in the first two rows of Table 8.

Limited information can be obtained on the abundance of Li from the observed *absence* of the Li I resonance lines at λ6707. To equation (9) we may add:

$$\text{from } \lambda 6707\text{: } \log N(\text{Li I}) = \log I + 1.74 + 1.84\,\theta + \text{const.}, \tag{10}$$

where I is the upper limit on the *observed* intensity of λ6707. We estimate this upper limit as follows. The [O I] line λ6300 appears very weakly on the same spectrograms on which λ6707 is absent. For λ6300 we measure $\log I = -1.96$, and we estimate that if this line were weaker by a factor of 2 it would not be detectable. That this is so is supported by the fact that the companion line λ6363 of [O I], for which the transition probability is smaller by a factor of 3, is not detected in emission on our spectrograms. We therefore estimate that for λ6707 the upper limit on log I is −2.0 in which case equation (10) becomes

$$\log N(\text{Li I}) < -0.26 + 1.84\,\theta + \text{const.} \tag{11}$$

By pairing this equation with each of equations (9) we obtain the results in the last four rows of Table 8. It appears that for any θ greater than the lower limit derived above for alkali metals, the ratios of Li I and K I to Na I are smaller in the comet than the corresponding atomic abundance ratios in the Sun, meteorites, or Earth's crust (Aller 1961). However, the range of possible Li/K ratios extends from the terrestrial to the solar value. The calculated ratio depends on the adopted temperature and on whether the

TABLE 8

APPARENT ATOMIC ABUNDANCES OF ALKALI METALS

	Lines Used	Comet 1965f			Abundance Ratios in		
		θ=2 0	θ=2 5	θ=3 0	Sun	Chondritic Meteorites	Earth's Crust
log]N(K I)/N(Na I)]	λ4044 λ3302	−2 6	−2 9	−3 3	−1 6	−1 1	−0 3
	λ7699 λ5890	−2 8	−3 1	−3 3			
log [N(Li I)/N(K I)]	λ4044 λ6707	<−2 1	<−2 7	<−3 3	−3 7	−1 5	−2 2
	λ7699 λ6707	<−1 8	<−1 7	<−1 6			
log [N(Li I)/N(Na I)]	λ3302 λ6707	<−4 7	<−5 6	<−6 6	−5 3	−2 6	−2 4
	λ5890 λ6707	<−4 6	<−4 8	<−4 9			

(non-existent) Li line is paired with λ4044 or with λ7699. The former line is itself very weak and lies in a spectral region remote from λ6707. On the other hand λ7699 is very strong, possibly self-absorbed, and is referred to a weakly exposed sky continuum. It should be recalled that it is not possible to decide whether or not λ7699 is self-absorbed from the intensity ratio λ7665/λ7699 because of telluric O_2 absorption at λ7665.

In addition to the above complications it is necessary to consider differential ionization of Li I, Na I, and K I. Since no element was observed in two stages of ionization on October 22, we cannot determine the state of ionization of the cometary gas by direct appeal to observation, although the decrease in the ratio Ca II λ3968/Fe I λ3969 between October 21.4 and October 22.7 (cf. Fig. 2 of this paper with Fig. 1 of Dufay *et al.* 1965) suggests that a marked decline in the degree of ionization of Ca II occurred after perihelion. However, we can estimate the lifetimes t_{ph} of neutral alkali metals against photo-ionization by solar radiation from

$$\frac{1}{t_{\text{ph}}} = \frac{\pi R_{\odot}^2}{4r^2hc}\int_0^{\lambda_T} \lambda F_\lambda^{\odot}\sigma_\lambda d\lambda\,, \tag{12}$$

where $R_{\odot}^2/4r^2 = 2.78 \times 10^{-4}$ is the geometrical dilution factor for solar radiation at the comet on October 22.7, $F_{\lambda}^{\odot}$ is the mean specific intensity of the solar disk, σ_{λ} is the cross-section for photo-ionization from the ground state, and λ_T is the threshold wavelength. Recent experimental cross-sections for Li I, Na I, and K I by Hudson (1964) and Hudson and Carter (1965*a*, *b*) and values of $F_{\lambda}^{\odot}$ given by Allen (1963, chap. ix) lead to the results given below:

Element	t_{ph} (sec)
Li I . . .	0 4 $\times 10^3$
Na I	6 7 $\times 10^3$
K I . .	. 4 0 $\times 10^3$

The small t_{ph} for Li I is due in part to its large cross-section at threshold and in part to the fact that the cross-section for Li I actually increases initially with decreasing wavelength below threshold, while those for Na I and K I go through deep minima approximately 200 Å shortward of their thresholds. We also have calculated lifetimes against collisional ionization by coronal electrons from

$$\frac{1}{t_{\text{coll}}} = n_e \int_{v_T}^{\infty} v f_v Q_v d v \,, \tag{13}$$

where n_e is the coronal electron density at the comet, Q_v is the collisional ionization cross-section as given by Allen (1963, chap. iii), f_v is the Maxwellian distribution of velocities v for the coronal temperature, and v_T is the threshold velocity for collisional ionization. Values of $n_e = 2 \times 10^3$ cm^{-3} (Pottasch 1960) and temperatures of 1.0×10^6 ° K and 2.0×10^6 ° K lead to the following results:

Element	t_{coll} (sec)	
	$T = 1\times10^6$ ° K	$T = 2\times10^6$ ° K
Li I	4 2$\times10^5$	2 8$\times10^5$
Na I	3 8$\times10^5$	2.5$\times10^5$
K I	2 4$\times10^5$	1 7$\times10^5$

Collisional ionization is unimportant relative to photo-ionization unless our estimates of n_e and T are wrong by orders of magnitude.

The radius of the nuclear region (from Fig. 5 where 1″ corresponds to 700 km) is about 3000 km. Since the sharpness of the cometary emission lines precludes internal motions greater than about 3 km/sec, the time spent by atoms in the nuclear region must be $t_{\text{nuc}} \geq 10^3$ sec. The rough equality $t_{\text{nuc}} \sim t_{\text{ph}}$ suggests that the dimensions of the region in which the Na I and K I emission occurs may be controlled to a considerable extent by the photo-ionization rates, and conversely that near the center of the nucleus these elements are mostly neutral. If this argument is qualitatively correct, it can be inverted to obtain an estimate of the expansion velocity in the nuclear region. The probability that an atom will move a distance d at a uniform expansion velocity V before being ionized is exp $(-d/Vt_{\text{ph}})$. If, as deduced in § *Vg* below, half of the neutral Na and K is within $d_{1/2} = 3000$ km of the nucleus, then $V = 1.44\, d_{1/2}/t_{\text{ph}} = 0.9$ km/sec.

Interesting as the Li content of a comet may be in connection with the chemical composition of a hypothetical, primordial solar nebula, we must draw the following discouraging conclusions: (1) While the upper limit on the Li I/Na I ratio appears to be low in the comet, so does the K I/Na I ratio. Our calculations indicate that the former ratio is sensitive to the effects of photo-ionization while the latter is not. Furthermore, the upper limits on the Li I/Na I and Li I/K I ratios cannot be determined unambiguously and are subject to considerable errors of observation as well as of analysis. (2) Even if the above difficulties could be surmounted, allowance must be made for molecular combinations. The totality of peculiar chemical affinities on the part of Li relative to Na and K is beyond the writer's grasp. A number of such peculiarities are described by Sidgwick (1950).

f) Apparent Abundances of Other Metals

Estimates of the apparent abundances of Cr I, Mn I, Ni I, and Cu I relative to Fe I are collected in Table 9. They were derived on the assumption that all five elements were distributed among their energy levels in Boltzmann distributions with $\theta = 1.11$. As for Na I and K I we *provisionally assume* that the ratios of neutral species are not affected by differential ionization in the nuclear region. The results do not particularly resemble those for any particular solar-system material. The Mn/Fe ratio is based on only one weak line, but the mere presence of the Mn emission line is incompatible with the discrepancy relative to the Sun and meteorites. The Ni/Fe ratio seems to be more like that of chrondrites than that of the Sun. The Cu abundance is unaccountably large.

The absence of lines of Mg I, Al I, Si I, Ti I, V I, and Co I lead to upper limits on their abundances relative to Fe I as indicated in Table 10. When allowance is made for errors

TABLE 9

ABUNDANCES OF Cr, Mn, Ni, AND Cu RELATIVE TO Fe

Element	log N(Neutral Metal)/N(Fe I) (Comet 1965f)	log N(Metal)/N(Fe)		
		Sun	Chondritic Meteorites	Earth's Crust
Cr	−1 9	−1 4	−1 9	−2 7
Mn	−1 2	−1 7	−1 9	−1 7
Ni	−1 1	−0 7	−1 3	−3 1
Cu	−0 2	−1 5	−3 5	−3 0

TABLE 10

UPPER LIMITS ON ABUNDANCES DERIVED FROM NON-APPEARANCE OF CERTAIN EMISSION LINES IN COMET 1965f

Element	Lines Used	log N(Neutral Element)/N(Fe I) (Comet 1965f)	log N(Element)/N(Fe I)		
			Sun	Chondritic Meteorites	Earth's Crust
Mg	λ3838 29	<−0 1	+0 8	+0 2	0 0
Al	λ3961 52	<−1 8	−0 4	−0 8	+0 5
Si	λ3905 53	<+0 6	+0 9	+0 2	+1 0
Ti	{λ3653 50, λ3998 64}	<−2 3	−1 9	−2 4	−1 0
V	{λ3703 58, λ4111 78}	<−2 4	−2 9	−3 4	−2 6
Co	λ3453 51	<−2 0	−1 9	−2 5	−3 4

of a factor of 2 in the upper limits and for the spread in abundances in other solar-system objects, it appears that only Al I, and possibly Mg I, can be regarded as abnormal. Since the photo-ionization cross-section for Al I exceeds that for Na I by a factor of 150 according to Burgess, Field, and Michie (1960), the absence of Al I lines probably can be explained by the short lifetime of Al I against photo-ionization.

Finally, we caution against comparison of the abundances of the alkali metals with those of Cr, Mn, Fe, Ni, and Cu until the reason for the different spatial distributions of these two groups of elements in the nuclear region of the comet is found. It is tempting to attribute the relatively narrow brightness distribution for Fe I to a high photo-ionization rate, but we are not aware of reliable cross-section data with which to test this notion.

g) *Surface Densities*

The quantity C in equation (6) may be written in terms of observed quantities as

$$C = F_\lambda^\odot \left(\frac{F_\lambda^{\text{sky}}}{F_\lambda^\odot}\right), \tag{14}$$

where $F_\lambda^\odot$ is the mean surface brightness of the solar disk and F_λ^{sky} is the surface brightness of the sky at the position of the comet. Values of $F_\lambda^\odot$ from Tables 2 and 5 of Minnaert (1953) for several wavelengths of interest are as follows:

λ(Å)	$F_\lambda^\odot$*	For Lines of
3250....	1 65	Cu I
3800 ..	3.24	Fe I, Ni I, Cr I
5900 ..	2 76	Na I
7700 ..	1 81	K I

* Unit = 10^6 ergs cm^{-2} Å^{-1} sterad^{-1}

For the brightness of the solar aureole in units of $F_\lambda^\odot$ at a distance of 7°.5 from the Sun we adopt a rough average value of 2.0×10^{-5} from observations of Abbot (1922), Newkirk and Eddy (1964), and the compilation by van de Hulst (1947). Then equation (6) becomes

$$\log N_0 = 16.98 + \log C(\lambda) + \log u + \log f(\lambda) + \log \lambda^3 I - \log gf + \theta\chi . \tag{15}$$

From the curve of growth in Figure 8 we then obtain for Fe I in the central region of the cometary nucleus

$$N(\text{Fe I}) = 3.4 \times 10^{10}$$

and for Na I

$$N(\text{Na I}) \geq 4.9 \times 10^{12} .$$

The lower limit for Na results from the lower limit on θ obtained in § V*e*. The CN (0,0) emission lines are comparable to medium strong Fe I emission lines on October 22 while the CN transition probabilities are 10–100 times smaller (Schadee 1964). Therefore, we expect that $N(\text{CN}) \sim 10^{12}$. If surface brightness is proportional to surface density, then the distributions in Figure 5 can be integrated with the assumption of spherical symmetry to obtain total numbers of emitting atoms

$$\mathfrak{N}(\text{Fe I}) = 7 \times 10^{27} \quad \text{and} \quad \mathfrak{N}(\text{Na I}) \geq 2 \times 10^{29} .$$

Half of the emitting atoms lie within about 2000 and 3000 km of the nucleus for Fe I and Na I, respectively. The volume densities in these regions are, therefore,

$$n(\text{Fe I}) = 180 \text{ cm}^{-3} \quad \text{and} \quad n(\text{Na I}) \geq 8000 \text{ cm}^{-3} .$$

From estimates of the likely contributions due to CN and other molecules (see Table 17 of Arpigny 1965) it seems unlikely that the average total density of atoms and molecules in the nuclear region exceeds 10^5 cm^{-3}. Therefore, the mean free path against excitation collisions will be greater than 3000 km for collision cross-sections smaller than about 7×10^{-14} cm^2. Since excitation cross-sections generally fall well below this limit, collisions are probably rare as was assumed in § V.

VI. THE MOLECULAR SPECTRUM

a) Description of the Observations

The first Lick spectrogram of comet 1965f was obtained in the dawn twilight of October 8, 1965, with the 120-inch coudé spectrograph (dispersion 16 Å/mm). The comet spectrum, contaminated by an appreciable sky spectrum, contains emissions due to the CN (0,0) and (0,1) bands, the C_2 (1,0), (2,1), (3,2), and (4,3) bands, weak CH features at λ4304 and λ4313, and very weak C_3 features near λ4050. On the following morning a much weaker 8 Å/mm spectrogram, free of contamination by the sky, shows only the stronger CN and CH features. On October 22 the only observed molecular features were those of the CN (0,0) and (1,1) bands. The latter was very weak and was found only by careful spectrophotometry. Crossley slit spectrograms (dispersion 430 Å/mm at Hγ) obtained on October 28 and 29 show only the bands of CN and C_2. While the C_2 (0,0) band at λ5165 was not present on spectrograms of October 22, it was so strong on October 28 and 29 that it is visible on exposures obtained with the IIaO emulsion. Thus the relative intensities of the CN and C_2 bands appear to have altered markedly between October 22 and October 28. However, the energy distributions of the sky at the two positions of the comet may affect the apparent relative intensities. The transition from neutral to Rayleigh scattering between October 22 and 28 that accompanies increasing angular distance from the Sun is in the correct sense to account for part of the apparent variation. Finally, a 16 Å/mm coudé spectrogram obtained on November 3 is very similar to that of October 8, except that it is virtually free of sky contamination. From the above observations it appears that, at solar distances of 0.4–0.6 a.u., the spectrum of comet 1965f resembled that of many previously observed comets and that the close approach to the Sun did not alter its spectroscopic characteristics. The absence of CN emission on spectrograms of October 13 reported by Dufay *et al.* (1965) is strange in view of our positive observations at earlier and later dates.

b) Intensities in the CN (0,0) Band

Intensities of lines in the CN (0,0) band derived from microphotometer tracings are given in Table 11. Intensities are given only for the nuclear region. On the more heavily exposed spectrograms intensities were measured as a function of position along the slit, but no certain evidence was found for internal motions such as that described by Greenstein (1958) for comet Mrkos (1957d). Therefore we have described the intensity distributions of the CN lines on October 8 and November 3 by the smooth curves in Figure 13 drawn through the normalized intensities of the lines $R(8)$–$R(16)$. It should be remarked that on all spectrograms except that of October 22 the molecular emission centroids are displaced westward relative to the cometary continuum. This effect is particularly noticeable on the Crossley spectrograms of October 28 and 29.

The half-intensity widths of the distributions in Figure 13 are about 30″ or 3 times greater than the value for October 22 (see Fig. 5). This variation in size of the emitting region lies well within the range of theoretical possibilities considered by Arpigny (1965)

TABLE 11

RELATIVE INTENSITIES OF CN LINES*

K	I[R(K)] Oct 8	Oct 9	Oct 22	Nov 3	K	I[P(K)] Oct 8	Oct 9	Oct 22	Nov 3
1.	.	. .	0 11	0 86	.	.	. ..		. . .
2	. ..	. ..	09	1 01	2 .		0 75	. ..	
3	1 17	1 05	16	1 20	3		1 48	0 19	
4	1 53	1 48	20	0 94	4		0 60	.19	
5	2 03	2 05	.33	0 47	5		1 30	17	
6.	1 80	1 84	34	1 15	6		2 32	24	
7	2 00	1 43	23	1 96	7		2 66	27	
8	1 67	1 22	17	1 79	8		1 82	.34	
9	1 50	1 27	34	2 58	9		1 51	24	
10	1 98	2 22	37	2 60	10 .		1 08	21	
11	2 76	3.10	26	2 75	11 .	...	1 39	35	
12	2 64	2 08	50	3 50	12		1 74	.42	
13	2 51	2 22	26	3 42	13 ..		3 35	26	
14	1 98	1 80	34	2 40	14 .	..	2 70	54	
15	2 58	2 40	33	2 05	15 .		2 22	.28	.
16	3 36	3 01	45	1 38	16		1 73	41	
17	2 70	2 29	17	1 50	17		2 28	38	
18	1 17	0 96	16	0 33	18 .		2 85	.51	.
19	2 15	1 55	15	0 62	19 .		2 56	.21	.
20.	1 74	0 96	26	0 64	20		1 26	19	
21	1 47	1 17	12	0 84	21 . .		. ..	24	
22	1 20	0 96	23	0 3:	22			0 33	
23 .	0 9:	. .	0 14	0 4:					
24	1 0:	.	. . .	0 4:					
25	0 9:			0 4:					
26	1 0:			0 3:					

* Intensities of CN lines for Oct 22 can be converted to the scale for metal lines in Table 2 by multiplying the CN intensities by the factor 0 054

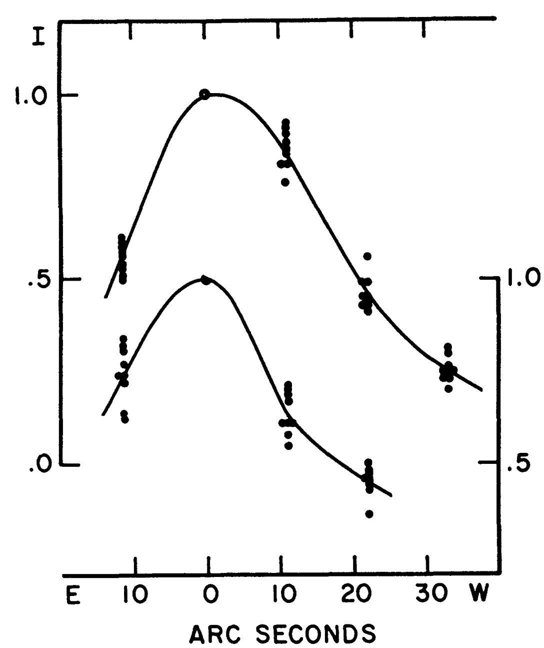

FIG. 13.—Surface brightness distributions of comet Ikeya-Seki derived from CN (0,0) emission lines on October 8 (*below*) and November 3 (*above*). Filled circles denote intensities of individual R-branch lines in units of their intensities at $\alpha = 0$.

The distance of the comet from Earth happened to be very nearly 1 a.u. for all the Lick spectroscopic observations so that 1″ corresponds to about 700 km in all cases.

The relative intensities of the R-branch lines on October 9, October 22, and November 3, taken from Table 11 are plotted in Figure 14. All three distributions show the mutilating effects due to fluorescence excited by solar radiation (Swings 1941). That the irregularities in these distributions are not due primarily to errors of observation or measurement is indicated by the good correlation in Figure 15 between the emission per unit line strength of the $R(K)$ and $P(K+2)$ lines on the spectrograms of October 9 and October 22.

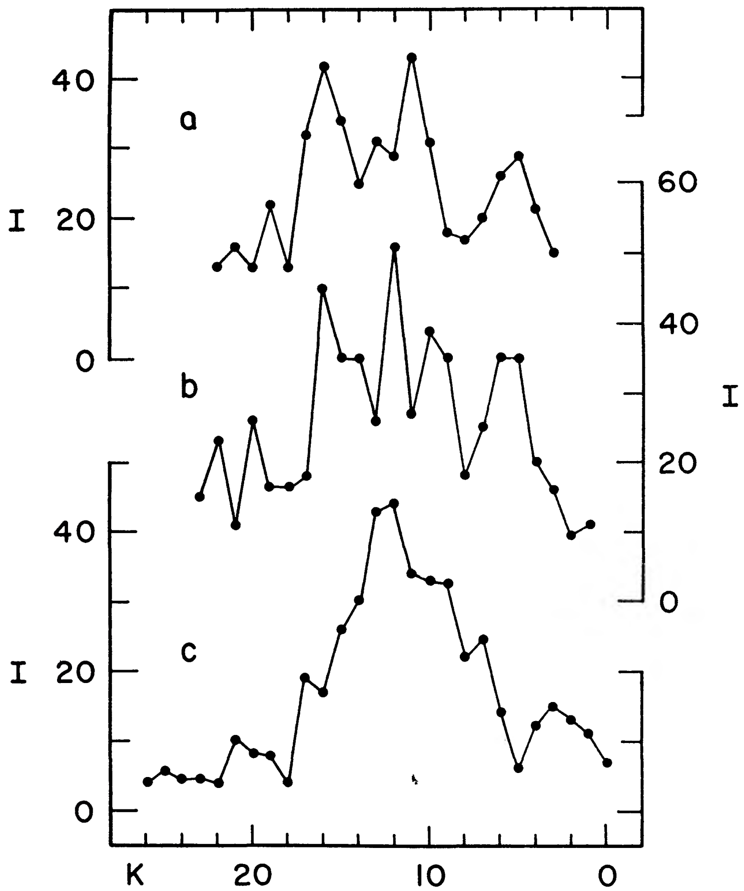

FIG 14 —Relative intensities of the $R(K)$ lines of the CN (0,0) band on (a) October 8, (b) October 22, and (c) November 3.

We can infer the relative populations of the rotational levels of the ground term of CN on the assumption that the observed emission is due to scattered solar radiation. Then, following Swings (1941), we write

$$I_R + I_P \propto N_{K-1} F_R{}^{\odot} \frac{S_{R(K-1)}}{g_{K-1}} + N_{K+1} F_P{}^{\odot} \frac{S_{P(K+1)}}{g_{K+1}}, \tag{16}$$

where I_P and I_R are the intensities of the $R(K-1)$ and $P(K+1)$ transitions, S_R and S_P are the strengths of these transitions, the g's are statistical weights, $F_R{}^{\odot}$ and $F_P{}^{\odot}$ are the Doppler-shifted monochromatic solar intensities observed at the comet at the wavelengths of the $R(K-1)$ and $P(K+1)$ transitions, and K is the rotation quantum number in the ground vibrational level. We replace N_{K-1} and N_{K+1} by N_K, use $S_R = K+1$, $S_P = K$, $g_K = 2K+1$ and define

$$I_{PR} = I_P + I_R \quad \text{and} \quad F_{PR} = F_R{}^{\odot}\frac{K}{2K-1} + F_P{}^{\odot}\frac{K+1}{2K+3}, \tag{17}$$

to obtain, approximately,

$$N_K \propto \frac{I_{PR}}{F_{PR}}. \tag{18}$$

The relative populations inferred from equation (18) are plotted in Figure 16. Formal rotational temperatures can be calculated from these empirical relative populations on the further assumption that

$$N_K \propto (2K+1)\exp\left[-\frac{BK(K+1)}{0.7T}\right], \tag{19}$$

in which case the slope fitted to the data for October 22 as in Figure 17 leads to a temperature of 760° K, very nearly the value derived from McKellar's (1944) expression

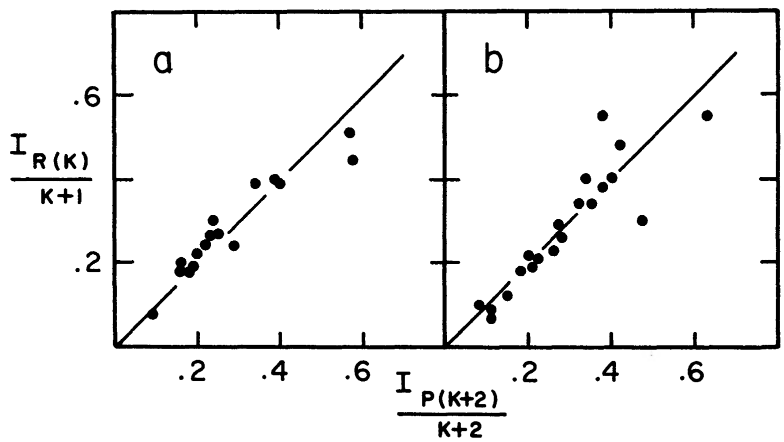

FIG. 15.—The correlation between intensity per unit line strength for the $R(K)$ and $P(K+2)$ lines of the CN (0,0) band on (*a*) October 9 and (*b*) October 22.

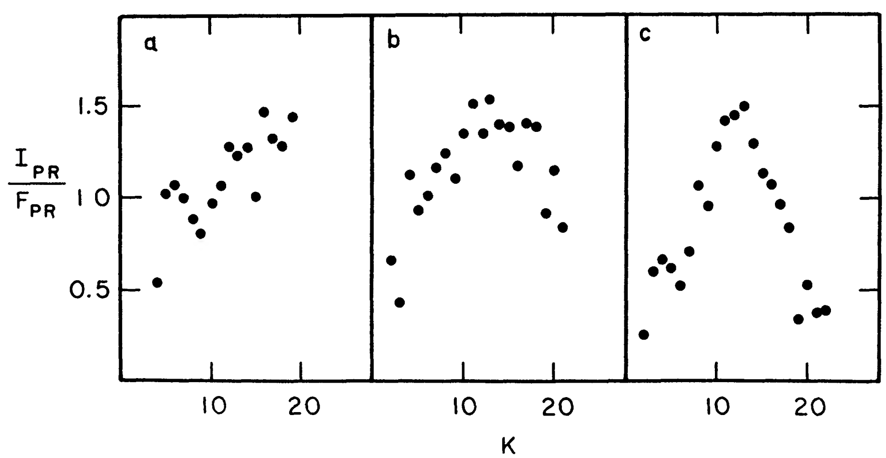

FIG. 16.—Relative populations of rotational levels in the ground vibrational state of CN inferred by the method described in § IV. Panels are labeled as for Fig. 14.

$T = 300r^{-1/2}$ for the solar distance of the comet on October 22.7. The scatter about the mean relation for October 22 is hardly larger than could be expected from observational errors. The populations inferred from the observations of October 9 and November 3 lead to formal rotational temperatures that are lower and higher than 760° K, respectively (the solar distances of the comet were nearly equal on these dates), and are not well represented by a Boltzmann distribution. Alternatively, we may compare our empirical populations with Arpigny's (1965) steady-state theory which gives the distribution in the rotational levels as a function of solar distance. A fair match between theory and observation is possible for November 3 if the theoretical distribution for

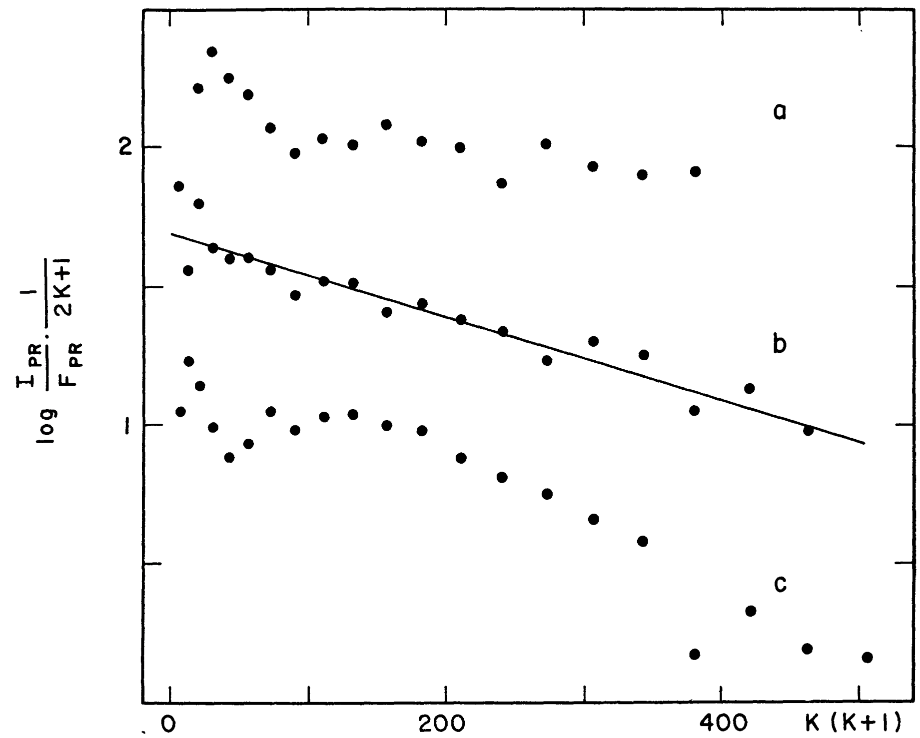

FIG. 17.—Plots of log $[(I_{PR}/F_{PR}\odot)(1/2K+1)]$ versus $K(K+1)$ for the observations of (*a*) October 9, (*b*) October 22, and (*c*) November 3, 1965. The formal rotational temperature of CN is determined by the slope of a straight line fitted to the observations.

$r = 0.75$ (as compared with the actual value of 0.65) is used. However, the observations of October 22 require the use of a distribution for $r = 0.4$ to 0.5, much greater than the actual value, and the data for October 9 can hardly be represented by any of Arpigny's distributions. The difficulty in the last case is due to the strength of the *R*- and *P*-branch lines for *K* greater than about 17. Since the spectrogram of October 8 is very similar to that of October 9 in this respect, the discrepancy appears to be a real one.

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